

COMMUNITY-BASED PROJECT LEARNING

CivicLight

A Streetlight Fault Reporting System for Citizens

[Computer Science And Engineering] · [Vijaya Vittala Institute Of Technology] · [2026-27]

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Abstract

CivicLight is a web-based platform that lets residents report malfunctioning streetlights by submitting a photograph and their precise GPS location in a few taps. Reports are organised into trackable tickets and surfaced on a live administrator dashboard, where municipal staff can monitor every open complaint, update its status, and close the loop with the citizen who raised it.

The system replaces slow, untracked phone or in-person complaints with a transparent, accountable digital workflow — improving night-time public safety and citizen trust in civic maintenance.

Introduction

Streetlights are a basic civic utility — when they fail, neighbourhoods lose visibility, safety, and a sense of being looked after. Today, reporting a faulty light usually means a phone call or a visit to the ward office, with little way to know if, or when, it will be fixed.



Night Safety

Dark streets raise the risk of accidents and crime.



Civic Trust

Visible follow-through builds confidence in local government.



Faster Response

Exact location and photo evidence speed up repairs.

Problem Statement

"Develop an application for citizens to report faulty streetlights using photographs and GPS locations."

- No structured channel exists for citizens to submit complaints with photographic evidence
- Citizens have no way to track the status of a complaint once it is filed
- Administrators lack a centralised, real-time view of all open faults
- Repairs are often delayed by vague or missing location information

Objectives

01 Enable photo + GPS based fault reporting for every citizen

03 Centralise every complaint on one live administrator dashboard

05 Improve transparency and accountability in civic maintenance

02 Provide secure, role-based logins for citizens and administrators

04 Track complaint status from submission through to resolution

06 Reduce the average turnaround time for streetlight repairs

Proposed System



Every new report syncs to the administrator dashboard instantly, and every status update syncs back to the citizen's own tracker — closing the loop without either side needing to refresh or follow up manually.

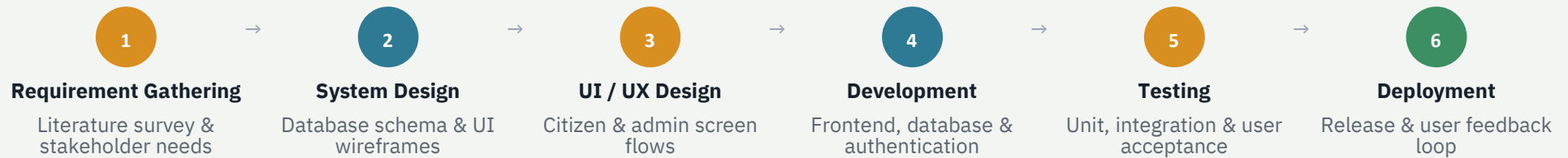
Hardware Requirements

Component	Purpose
Citizen's smartphone or PC (camera + GPS)	Capture photo evidence and location of the fault
Administrator's desktop or laptop	Access and manage the live complaints dashboard
Internet connectivity	Required for both citizens and administrators
Cloud server infrastructure	Hosts the application, database and uploaded photos

Software Requirements

Layer	Technology
Frontend	HTML5, CSS3, JavaScript (ES6)
Data layer	Real-time NoSQL cloud document database
Media storage	Cloud object storage for uploaded photographs
Browser APIs	Geolocation API, File / Camera API
Authentication	Role-based email & password login (Citizen / Administrator)
Client software	Modern web browser — Chrome, Firefox or Edge

Methodology



An iterative, agile-inspired approach was followed, with each stage reviewed before moving to the next so that citizen and administrator needs stayed central to every design decision.

Screenshots of Results/Outcomes

Citizen Dashboard

7
Total reports

2
In progress

4
Resolved



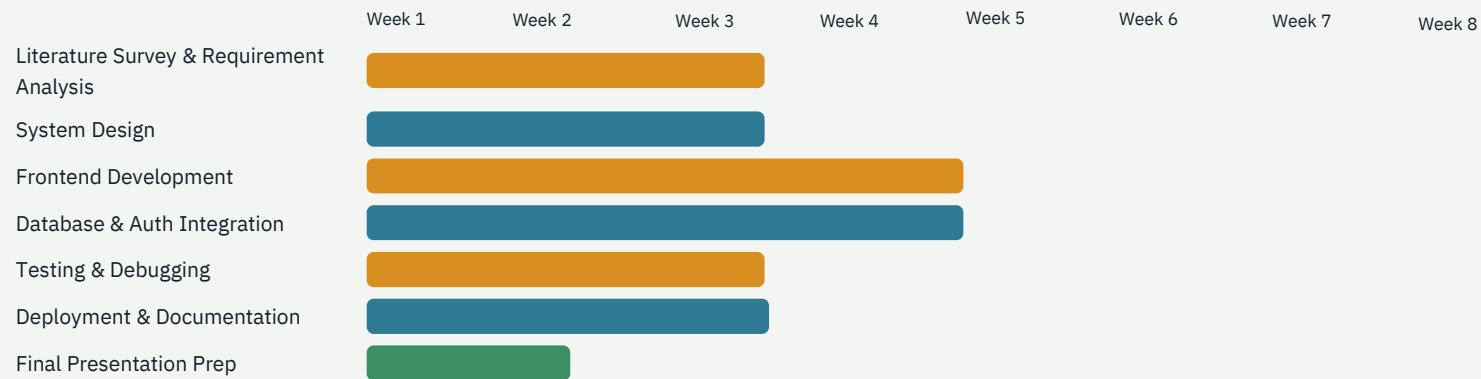
CL-2026-9F3A • Flickering light
Opposite Ganesha temple, 4th Main Road

Administrator Dashboard

Ticket	Status
CL-2026-9F3A	In Progress
CL-2026-6B1D	Submitted
CL-2026-2A7E	Resolved

Preview of the live application interface — citizen ticket tracking (left) and the real-time administrator complaint table (right).

Project Timeline — Gantt Chart



Indicative 8-week schedule for the Community-Based Project Learning cycle.

Conclusion

CivicLight successfully bridges the gap between citizens and municipal administrators for streetlight maintenance. By combining photo evidence, GPS geotagging and a live tracking workflow, it replaces an opaque, manual process with one that is fast, transparent and accountable — directly improving night-time safety across a ward.

Future Scope

- SMS / email alerts to citizens at each status change
- Analytics dashboard to identify recurring fault hotspots
- Integration with IoT light sensors for automatic fault detection

References

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